Chapter 8
Solar Radiation Dynamics

Solar radiation (insolation) is the primary driving force for Earth's atmospheric, biophysical, and hydrologic processes. Knowing the amount of radiation at different geographic locations at different times is therefore necessary in many fields including energy production, agriculture, meteorology, ecology, and urban planning. We modeled direct solar radiation and cast shadows in an urban setting to show how different spatial configurations of buildings change the amount of sunlight available throughout the day and the year.

8.1 Solar Radiation Modeling

Solar radiation modeling is critical for the optimal site selection of solar power plants (Carrion et al. 2008; Janke 2010) and for placing photovoltaic systems in often complex urban environments (Freitas et al. 2015; Hofierka and Kaňuk 2009; Jakubiec and Reinhart 2013). In urban design dynamic solar radiation simulations are used to study the effect of urban geometry and building configurations on solar access and shading conditions in order to get insight into the "urban canyon" phenomenon (Arnfield 1990; Lobaccaro and Frontini 2014).

On a global scale solar radiation depends spatially and temporally on the orientation of the Earth relative to the Sun. At a local scale solar radiation is influenced by topography (i.e. an elevation surface's inclination, orientation, and shadows), atmospheric conditions, and surface properties such as land cover. We can analyze the spatial distribution and dynamics of solar radiation and cast shadows using the equations that relate the sun position to the geometry of the terrain surface (Hofierka and Suri 2002). The clear-sky solar radiation model applied in this chapter is based on work undertaken for the development of the European Solar Radiation Atlas (Page et al. 2001; Rigollier et al. 2000; Scharmer and Greif 2000). We include here only the equations for direct solar radiation—beam irradiance—which

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A. Petrasova et al., *Tangible Modeling with Open Source GIS*,
DOI 10.1007/978-3-319-25775-4_8

is influenced by the local topography. For a complete set of equations see Hofierka and Suri (2002). The diffuse and reflected components as well as atmospheric parameters can be also found in Rigollier et al. (2000).

First we compute position of the Sun in the sky as a function of time and location on Earth. The Sun declination angle depends on the day number (1–365 or 366):

$$\delta = \arcsin(0.3978 \sin(j' - 1.4 + 0.0355 \sin(j' - 0.0489))) \tag{8.1}$$

where δ is the Sun declination angle in radians and j' is expressed as day angle in radians:

$$j' = 360^\circ \cdot j/365.25, \quad j = 1, 2, \ldots, 365\,(366) \tag{8.2}$$

The position of the Sun in respect to a horizontal plane is defined by the solar altitude and solar azimuth. The solar altitude angle γ_s is a function of the solar hour angle ω, the solar declination angle δ and the latitude φ of the given location:

$$\sin \gamma_s = \cos \varphi \cos \delta \cos \omega + \sin \varphi \sin \delta \tag{8.3}$$

Given that the solar hour angle ω changes at 15° per hour and equals zero at noon it can be computed from local solar time t in decimal hours on the 24 h clock:

$$\omega = 15^\circ \cdot (t - 12) \tag{8.4}$$

The solar azimuth angle α_s (a horizontal angle between the sun and meridian measured from east) can be expressed as a function of the solar altitude angle γ_s, the solar declination angle δ, and the latitude φ of the given location:

$$\cos \alpha_s = (\sin \varphi \sin \gamma_s - \sin \delta)/ \cos \varphi \cos \gamma_s \tag{8.5}$$

$$\sin \alpha_s = \cos \delta \sin \omega / \cos \gamma_s \tag{8.6}$$

The position of the Sun in respect to an inclined plane with slope β and aspect α (e.g., a hillslope or a building roof) is described by a solar incidence angle ν:

$$\cos \nu = \cos \gamma_s \sin \beta \cos(\alpha_s - \alpha) + \sin \gamma_s \cos \beta \tag{8.7}$$

To estimate the duration of solar radiation we compute the sunrise and sunset angle $\omega_{r,s}$ over a horizontal plane:

$$\cos \omega_{r,s} = -\tan \varphi \tan \delta \tag{8.8}$$

and the corresponding time of sunrise T_{sr} and sunset T_{ss} in hours is then:

$$T_{sr} = 12 - \omega_{r,s}/15^\circ \qquad T_{ss} = 12 + \omega_{r,s}/15^\circ \tag{8.9}$$

The sunrise/sunset angle over a south-facing inclined plane (for a location in the northern hemisphere) is computed as follows:

$$\cos\omega'_{r,s} = -\tan(\varphi - \beta)\tan\delta, \qquad \omega''_{r,s} = \min(\omega_{r,s}, \omega'_{r,s}) \tag{8.10}$$

where $\omega''_{r,s}$ is the sunrise and sunset angle over the inclined plane. More general equations can be found in Hofierka and Suri (2002).

Solar irradiation analysis has been implemented in GRASS GIS by Hofierka and Suri (2002) in the module *r.sun*. In addition to the direct solar irradiation described above the module incorporates diffuse and reflected irradiation and cast shadows. The module *r.sun* works in two modes. In the first mode it computes the solar incident angle in degrees and solar irradiance values in $\mathrm{W\,m^{-2}}$ for a specific instance in time. In the second mode it computes the daily sum of solar irradiation in $\mathrm{Wh\,m^{-2}\,day^{-1}}$ and the duration of the direct irradiation. To compute a series of direct irradiance maps for a given day with a selected time interval we use the add-on *r.sun.hourly*, which calls *r.sun* in a convenient loop. Similarly we use the add-on *r.sun.daily* to compute daily sums of direct irradiation for the entire year.

8.2 Case Study: Solar Irradiation in Urban Environment

In this case study we studied how different spatial configuration of buildings changed direct solar radiation and cast shadows dynamics throughout the day. We placed wooden blocks of different shapes and sizes representing hypothetical buildings on a flat surface with an approximate scale of 1:300. We located this abstract site in Raleigh, North Carolina at latitude 36°. We examined two building configurations—one with four medium-sized houses with gabled roofs and the other with three medium-sized houses with gabled roofs and one tall building with a flat roof (Fig. 8.1a, b).

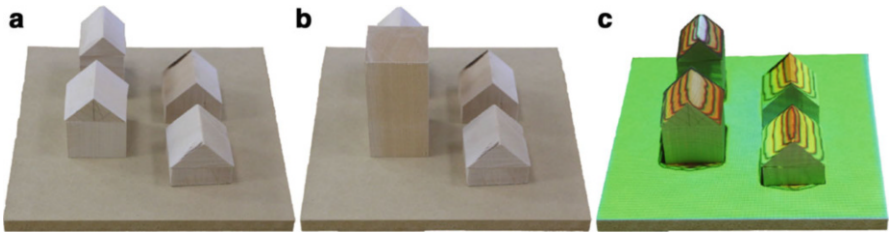

Fig. 8.1 Two configurations of buildings at approximately 1:300 scale viewed from the south. (**c**) Shows the DSM and contours derived from a scan and projected over buildings

8.2.1 The Impact of Building Configuration on Cast Shadows

We compared the shadows cast by the buildings in both configurations during summer and winter solstice. After scanning the scenes we used the add-on *r.sun.hourly* to derive binary raster time series of cast shadows from 6:00 to 22:00 (local solar time) with a 30-minute time step. We had to specify the selected date as a day of year number. For example if we pick the summer solstice in 2015 we can easily compute the day of year using the Python *datetime* module:

```
import datetime
datetime.datetime(2015, 6, 21).timetuple().tm_yday
```

which gives us the number 172.

Running the following commands results in a time series of binary rasters with zeros representing cast shadow based on the direct (beam) radiation. The time series is registered in a spatio-temporal dataset and can be easily rendered as an animation and projected over the physical model.

```
r.slope.aspect elevation=scan aspect=aspect slope=slope
r.sun.hourly -t -b elevation=scan aspect=aspect slope=slope \
    start_time=6 end_time=22 time_step=0.5 day=172 year=2015 \
    beam_rad_basename=shadows_summer
g.gui.animation strds=shadows_summer
```

By changing the day parameter to 356 we obtained the time series of cast shadows for the winter solstice. Figure 8.2 compares the cast shadows at 16:30 and 19:00 (local solar time) during the summer and winter solstice for the two different building configurations.

There is a clear difference in the length of cast shadows during winter and summer. The orientation of the cast shadows at the time close to sunset changes during the year as shown in Fig. 8.2a, c. During the summer the Sun sets below the horizon later (after 19:00) far to the northwest, while it sets earlier (after 16:30) and far to the southwest during winter. As a result the house in the southeast corner only has a view of the sunset during part of the year as the summer view is blocked by another building.

By adding a tall building to the scene we cast more shadow in the space between the houses and on houses in the east. Depending upon the season one of the houses on the east side becomes shadowed by the high building sooner than the other.

8.2.2 The Impact of Building Configuration on Direct Solar Irradiation

Cast shadows give us information about the availability of direct solar radiation at a particular time on particular days. However, we are often interested in the cumulative solar irradiation, which tells us the amount of solar energy received on

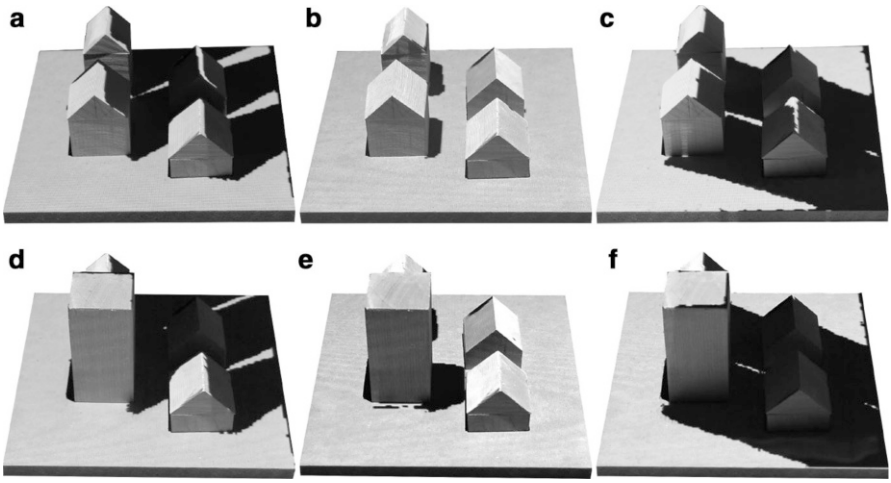

Fig. 8.2 A comparison of cast shadows viewed from south. The *first and second rows* show two different building configurations. The *first and second columns* show cast shadows at 16:30 local solar time during the winter (**a**), (**d**) and summer solstice (**b**), (**e**), respectively. The *third column* shows cast shadows at 19:00 local solar time during the summer solstice. A figure for winter solstice at 19:00 is not shown as it would be after sunset

a given surface during a given time interval. Such information can then be used to assess the suitability of a location for solar energy applications.

To compute direct solar irradiation for individual days we called the add-on *r.sun.daily* with parameters similar to those we used for *r.sun.hourly*. We specified the first and last day of the year resulting in 365 daily irradiation maps. Additionally by specifying the `beam_rad` parameter we computed the sum of all of the daily maps. Again we can animate the time series projected over the scene to get more insight into the spatial patterns of solar radiation over the course of a year.

```
r.slope.aspect elevation=scan aspect=aspect slope=slope
r.sun.daily -t elevation=scan aspect=aspect slope=slope \
    start_day=1 end_day=365 day_step=1 \
    beam_rad=radiation_year_sum \
    beam_rad_basename=radiation_day_sum
g.gui.animation strds=radiation_day_sum
```

Figure 8.3 shows direct solar irradiation in $\mathrm{Wh\,m^{-2}\,day^{-1}}$ during the winter solstice, spring equinox, and summer solstice with a common color ramp. The color ramp was specified in a plain text file `solar_color.txt` as:

```
100% red
70% yellow
0% gray
```

and then this color ramp was set for the entire time series:

```
t.rast.colors input=radiation_day_sum rules=solar_color.txt
```

Fig. 8.3 Direct solar irradiation in Wh m^{-2} day^{-1} during the winter solstice, spring equinox, and summer solstice

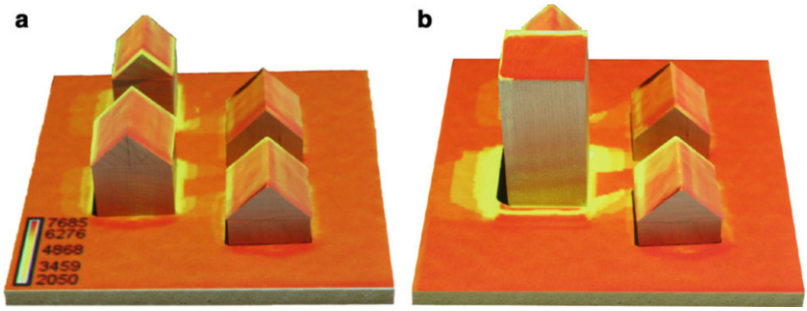

Fig. 8.4 Solar irradiation patterns in March for different building configurations

Figure 8.4a shows the solar irradiation on a particular day in March in order to compare the cumulative solar irradiation for the two building configurations. In the first scenario there is variable irradiation in the space between the buildings as a result of the interaction between the spatial configuration of the buildings and the dynamic solar geometry. In the second scenario the pattern is even more pronounced because the street gets less direct sunlight. The slope of the buildings' roofs affects the amount of irradiation as a visual comparison of the color of the flat and tilted roofs clearly demonstrates. Furthermore, the shadow cast by the tall building has lowered the irradiation values on part of the roof of the southeastern house.

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